

Mikkili Koushik

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Portfolio: [LINK](#)

PROFESSIONAL SUMMARY

B.Tech Computer Science student with two internship experiences and hands-on expertise in AI/ML, Computer Vision, and NLP application development. Developed real-time hand gesture control systems, multilingual speech recognition and translation solutions, and sentiment analysis applications using Python. Proficient in Python, PyTorch, OpenCV, MediaPipe, and NLTK, with strong foundations in Data Structures & Algorithms, machine learning, and agile software development. Passionate about building intelligent, user-centric applications that combine AI with real-world problem solving.

EDUCATION

B.Tech – Computer Science Engineering	2022 – 2026
<i>Dhanalakshmi Srinivasan University CGPA: 7.34 (up to Sem 7)</i>	
Higher Secondary Certificate (MPC)	2020 – 2022
<i>Narayana Junior College 55%</i>	
Secondary School Certificate (SSC)	2019 – 2020
<i>Narayana High School CGPA: 9.3</i>	

TECHNICAL SKILLS

Languages:	Python, java
Frameworks & Tools:	PyTorch, OpenCV, NLTK, pandas, matplotlib,
Database:	MySQL
Concepts:	OOP, Data Structures & Algorithms, Computer Vision, NLP, Machine Learning, Gesture Recognition
OS:	Linux, Windows
Soft Skills:	Problem Solving, Communication, Time Management

PROJECTS

Air Mouse – Hand Gesture Controller ([Github](#))

Python · OpenCV · MediaPipe Hand Landmarker · pyautogui · NumPy

Problem

- Touch-based mouse control is not suitable for contactless environments.
- Webcam hand tracking is unstable and causes cursor jitter.

Solution

- Built a real-time air mouse using MediaPipe hand tracking (21 landmarks, 30 FPS).
- Used NumPy smoothing for stable cursor movement.
- Added pinch gestures for left/right click with cooldown protection.
- Implemented scroll control based on hand position with conflict handling.
- Added volume control using open-hand and fist gestures.
- Enabled feature toggles (mouse/scroll/volume) with keyboard shortcuts and live overlay.

Speech Translation & Emotion Detector ([GitHub](#))

Python · SpeechRecognition · Vosk · langdetect · Hugging Face Transformers · BART

Problem

- Voice assistants need to understand multiple languages and process speech accurately.
- Long voice inputs are difficult to analyze and summarize.
- Emotion detection from speech text is needed for better user interaction.

Solution

- Developed a multilingual voice assistant that captures live speech and transcribes it using Google Speech Recognition and Vosk (offline support).
- Automatically detects language and translates speech to English for NLP processing.
- Performs keyword-based emotion detection (happy, sad, angry, neutral).
- Generates concise summaries of long speech inputs using the BART Transformer model.

AI Virtual Assistant for Desktop

([GitHub](#))

Python · Vosk · SpeechRecognition · pytsx3 · MediaPipe · OpenCV · pyautogui · pandas · matplotlib · tkinter · subprocess

Problem

- Voice assistants often fail without internet connectivity and provide limited system interaction.
- Users need a hands-free way to control the computer and analyze data through voice commands.
- Dataset analysis and visualization typically require technical knowledge and manual effort.

Solution

- Developed a dual-mode voice assistant that automatically switches between online and offline speech recognition based on network availability.
- Integrated an Air Mouse mode, allowing users to control the cursor, clicks, scrolling, and volume through hand gestures.
- Added text-to-speech responses and voice commands for web browsing, time, date, and system control.
- Implemented CSV data analysis with spoken summaries and automatic visualization using charts and graphs.

Sentiment Analysis Tool

([GitHub](#))

Python · NLTK · NLP · Internship Project @ Slash Mark IT Solutions

Problem

- E-commerce platforms receive large volumes of customer reviews, making manual sentiment analysis time-consuming and inefficient.
- Raw text data requires preprocessing before it can be used for sentiment classification.

Solution

- Developed an NLP pipeline to classify customer reviews as positive or negative sentiment.
- Applied NLTK-based preprocessing, including tokenization, stopword removal, and frequency-based feature extraction.
- Built the solution during an internship with a focus on production readiness and agile development practices.

INTERNSHIP EXPERIENCE

AI/ML Intern

May -Jun 2025

Micro Information Technology Services | Mullana, Ambala | Certificate ID: MITS136352

- Developed an offline multilingual speech recognition system using Vosk to process voice input without internet connectivity.
- Implemented automatic language detection for multilingual speech with approximately 75% detection accuracy.
- Built a speech translation pipeline that converts detected languages into English for further processing.
- Reduced dependency on cloud-based APIs by enabling fully offline speech processing.
- Tested and validated the system using Hindi, Telugu, and English speech samples to evaluate real-world performance.

Software Development Intern

Oct – Dec 2024

Slash Mark IT Solutions (OPC) Pvt. Ltd.

- Built a multilingual speech processing system using Google Speech Recognition and Vosk for online and offline transcription.
- Implemented automatic language detection and translation to English using langdetect.
- Developed keyword-based emotion analysis to identify user sentiment (happy, sad, angry, neutral).
- Integrated the Hugging Face BART model (facebook/bart-large-cnn) to summarize long speech inputs.
- Enabled support for multiple languages including Hindi, Telugu, and English for seamless voice interaction.

CERTIFICATIONS

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| • AI/ML Internship Certificate | <i>Micro Information Technology Services</i> |
| • Natural Language Processing | <i>Infosys</i> |
| • Python Programming | <i>Scaler</i> |

